

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3
```

```
In [2]: conn = sqlite3.connect('customer_churn.db')
sql_query = """
    SELECT name
    FROM sqlite_master
    where type = 'table'
    """
tables = pd.read_sql(sql_query, conn)
for table_name in tables['name']:
    df = pd.read_sql(f"SELECT * FROM {table_name}", conn)
    globals()[f"df_{table_name}"] = df
    print(f"Created dataframe: df_{table_name}")

conn.close()
```

Created dataframe: df_db_customer

Created dataframe: df_db_subscription

Created dataframe: df_db_support

```
In [3]: conn = sqlite3.connect('customer_churn.db')

for table_name in tables['name']:
    print(f"\nTable Name: {table_name}")
    # Get column information
    columns_query = f"PRAGMA table_info({table_name});"
    columns = pd.read_sql(columns_query, conn)
    print("Columns:")
    print(columns['name'].tolist())

conn.close()
```

Table Name: db_customer

Columns:

['customerid', 'name', 'country', 'state', 'gender', 'dob', 'interests', 'zipcode']

Table Name: db_subscription

Columns:

['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score']

Table Name: db_support

Columns:

['customerid', 'complaint_date', 'escalations', 'csat_score', 'col_1', 'comment']

```
In [4]: df_db_customer.head()
```

```
Out[4]:
```

	customerid	name	country	state	gender	dob	interests	pincode
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00	travel	None
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00	None	None
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00	movie	None
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00	None	None
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00	drama	None

```
In [5]: df_db_customer.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   customerid  21 non-null    object
1   name        21 non-null    object
2   country     18 non-null    object
3   state       21 non-null    object
4   gender      21 non-null    object
5   dob         21 non-null    object
6   interests   4 non-null     object
7   pincode     0 non-null     object
dtypes: object(8)
memory usage: 1.4+ KB
```

```
In [6]: # a. drop columns - intereset and pincode
# b. rename colums - name
# c. change datatype - dob--- obj to dob
# d. fixing missing values in country
# e. data standardisation - gender
```

```
In [7]: df_db_customer.tail()
```

Out[7]:

	customerid	name	country	state	gender	dob	interests	pincode
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00	None	None
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00	None	None
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00	None	None
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00	None	None
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00	None	None

```
In [8]: df_db_customer.rename(columns = {'name':'Customer_Name'},inplace=True)
df_db_customer
```

Out[8]:

	customerid	Customer_Name	country	state	gender	dob	interests	pincode
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00	travel	Nor
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00	None	Nor
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00	movie	Nor
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00	None	Nor
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00	drama	Nor
5	0013-MHZWF	durga	None	Delhi	Women	1988-12-10 00:00:00	None	Nor
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21 00:00:00	None	Nor
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14 00:00:00	None	Nor
8	0015-UOCOJ	maya	None	Kathmandu	Women	1985-07-07 00:00:00	None	Nor
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29 00:00:00	None	Nor
10	0017-DINOC	shiva	India	Maharashtra	Men	1997-01-22 00:00:00	None	Nor
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18 00:00:00	None	Nor
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01 00:00:00	None	Nor
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25 00:00:00	None	Nor

	customerid	Customer_Name	country	state	gender	dob	interests	pincode
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Men	1979-11-11 00:00:00	None	Nor
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28 00:00:00	job	Nor
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00	None	Nor
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00	None	Nor
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00	None	Nor
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00	None	Nor
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00	None	Nor

```
In [9]: df_db_customer.drop(df_db_customer.columns[-2:],axis=1,inplace=True)
```

```
In [10]: df_db_customer
```

Out[10]:

	customerid	Customer_Name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00
5	0013-MHZWF	durga	None	Delhi	Women	1988-12-10 00:00:00
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21 00:00:00
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14 00:00:00
8	0015-UOCOJ	maya	None	Kathmandu	Women	1985-07-07 00:00:00
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29 00:00:00
10	0017-DINOC	shiva	India	Maharashtra	Men	1997-01-22 00:00:00
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18 00:00:00
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01 00:00:00
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25 00:00:00
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Men	1979-11-11 00:00:00
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28 00:00:00
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00

In [11]: `# df_db_customer.drop(columns=['interests','pincode'],inplace=True)`

In [12]: `df_db_customer`

Out[12]:

	customerid	Customer_Name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00
5	0013-MHZWF	durga	None	Delhi	Women	1988-12-10 00:00:00
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21 00:00:00
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14 00:00:00
8	0015-UOCOJ	maya	None	Kathmandu	Women	1985-07-07 00:00:00
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29 00:00:00
10	0017-DINOC	shiva	India	Maharashtra	Men	1997-01-22 00:00:00
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18 00:00:00
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01 00:00:00
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25 00:00:00
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Men	1979-11-11 00:00:00
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28 00:00:00
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00

In [13]: `df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])`

In [14]: `df_db_customer['dob']`

```
Out[14]: 0    1982-04-12
         1    1995-11-23
         2    1978-02-15
         3    2001-08-30
         4    1990-05-05
         5    1988-12-10
         6    1976-09-21
         7    1999-03-14
         8    1985-07-07
         9    1993-10-29
        10    1997-01-22
        11    1981-06-18
        12    2004-12-01
        13    1992-04-25
        14    1979-11-11
        15    1986-02-28
        16    1994-08-19
        17    2000-09-02
        18    1983-12-30
        19    1991-05-14
        20    1977-10-06
        Name: dob, dtype: datetime64[ns]
```

```
In [15]: df_db_customer.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   customerid      21 non-null    object
 1   Customer_Name   21 non-null    object
 2   country         18 non-null    object
 3   state           21 non-null    object
 4   gender          21 non-null    object
 5   dob             21 non-null    datetime64[ns]
dtypes: datetime64[ns](1), object(5)
memory usage: 1.1+ KB
```

```
In [16]: df_db_customer['gender'].unique()
```

```
Out[16]: array(['Male', 'Female', 'Women', 'Men'], dtype=object)
```

```
In [17]: df_db_customer['gender']=df_db_customer['gender'].replace({'Men':'Male','Women':'Fe
```

```
In [18]: df_db_customer
```

Out[18]:

	customerid	Customer_Name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05
5	0013-MHZWF	durga	None	Delhi	Female	1988-12-10
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14
8	0015-UOCOJ	maya	None	Kathmandu	Female	1985-07-07
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29
10	0017-DINOC	shiva	India	Maharashtra	Male	1997-01-22
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Male	1979-11-11
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Male	1991-05-14
20	0023-UYUPN	sudevi	India	Maharashtra	Female	1977-10-06

In [19]:

df_db_customer

Out[19]:

	customerid	Customer_Name	country	state	gender	dob
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05
5	0013-MHZWF	durga	None	Delhi	Female	1988-12-10
6	0013-SMEOE	mina	India	Meghalaya	Female	1976-09-21
7	0014-BMAQU	madan	India	Rajasthan	Male	1999-03-14
8	0015-UOCOJ	maya	None	Kathmandu	Female	1985-07-07
9	0016-QLJIS	arjun	Nepal	Kathmandu	Male	1993-10-29
10	0017-DINOC	shiva	India	Maharashtra	Male	1997-01-22
11	0017-IUDMW	rangadevi	India	Karnataka	Female	1981-06-18
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01
13	0019-EFAEP	raju	India	Meghalaya	Female	1992-04-25
14	0019-GFNTW	Madhav	India	Uttar Pradesh	Male	1979-11-11
15	0020-INWCK	parvati	India	Delhi	Female	1986-02-28
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19
17	0021-IXXGC	vishakha	India	Rajasthan	Female	2000-09-02
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Male	1991-05-14
20	0023-UYUPN	sudevi	India	Maharashtra	Female	1977-10-06

In [20]: `df_db_customer[df_db_customer['country'].isna()]`

Out[20]:

	customerid	Customer_Name	country	state	gender	dob
5	0013-MHZWF	durga	None	Delhi	Female	1988-12-10
8	0015-UOCOJ	maya	None	Kathmandu	Female	1985-07-07
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01

In [21]: `df_db_customer[['country','state']]`

Out[21]:

	country	state
0	India	Maharashtra
1	India	Karnataka
2	India	Delhi
3	India	Nagaland
4	India	Delhi
5	None	Delhi
6	India	Meghalaya
7	India	Rajasthan
8	None	Kathmandu
9	Nepal	Kathmandu
10	India	Maharashtra
11	India	Karnataka
12	None	Telangana
13	India	Meghalaya
14	India	Uttar Pradesh
15	India	Delhi
16	India	Meghalaya
17	India	Rajasthan
18	India	Telangana
19	India	Uttar Pradesh
20	India	Maharashtra

```
In [22]: # Country amd state unique value pair
state_country_mapping = df_db_customer.dropna(subset=['country']).set_index('state')
df_db_customer['country'] = df_db_customer['country'].fillna(df_db_customer['state'])
```

```
In [23]: df_db_customer[df_db_customer['country'].isna()]
```

Out[23]:

customerid	Customer_Name	country	state	gender	dob
------------	---------------	---------	-------	--------	-----

```
In [24]: df_db_customer[['country', 'state']]
```

Out[24]:

	country	state
0	India	Maharashtra
1	India	Karnataka
2	India	Delhi
3	India	Nagaland
4	India	Delhi
5	India	Delhi
6	India	Meghalaya
7	India	Rajasthan
8	Nepal	Kathmandu
9	Nepal	Kathmandu
10	India	Maharashtra
11	India	Karnataka
12	India	Telangana
13	India	Meghalaya
14	India	Uttar Pradesh
15	India	Delhi
16	India	Meghalaya
17	India	Rajasthan
18	India	Telangana
19	India	Uttar Pradesh
20	India	Maharashtra

In [25]: `df_db_subscription.head()`

Out[25]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	A
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	A
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	M
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	A
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	M

```
In [26]: df_db_subscription.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerid            21 non-null    object
1   subscription_start_date 21 non-null    object
2   subscription_type      21 non-null    object
3   renewal_date          21 non-null    object
4   plan_type              21 non-null    object
5   contract_type          21 non-null    object
6   cancellation_date       6 non-null     object
7   cancellation_reason     6 non-null     object
8   monthly_charges        21 non-null    float64
9   cltv                   21 non-null    int64
10  churn_score             21 non-null    int64
dtypes: float64(1), int64(2), object(8)
memory usage: 1.9+ KB

In [27]: date_col = ['subscription_start_date','cancellation_date','renewal_date']
df_db_subscription[date_col]=df_db_subscription[date_col].apply(pd.to_datetime)
df_db_subscription[date_col]
```

Out[27]:

	subscription_start_date	cancellation_date	renewal_date
0	2021-03-15	NaT	2025-03-15
1	2020-08-01	2024-09-10	2024-08-01
2	2022-11-20	NaT	2025-11-20
3	2019-05-10	NaT	2025-05-10
4	2023-01-05	2024-02-28	2024-01-05
5	2022-06-18	NaT	2025-06-18
6	2021-09-30	2024-11-15	2024-09-30
7	2020-02-14	NaT	2025-02-14
8	2023-07-22	NaT	2024-07-22
9	2022-04-03	NaT	2025-04-03
10	2021-12-01	NaT	2025-12-01
11	2023-03-17	2024-05-01	2024-03-17
12	2020-10-09	NaT	2025-10-09
13	2022-08-25	2024-10-31	2024-08-25
14	2019-11-11	NaT	2025-11-11
15	2023-05-06	NaT	2025-05-06
16	2022-01-19	NaT	2024-01-19
17	2021-07-07	NaT	2025-07-07
18	2023-09-14	2024-09-14	2024-09-14
19	2020-06-23	NaT	2025-06-23
20	2022-12-31	NaT	2025-12-31

In [28]: `df_db_subscription.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customerid                           21 non-null     object
1   subscription_start_date               21 non-null     datetime64[ns]
2   subscription_type                     21 non-null     object
3   renewal_date                         21 non-null     datetime64[ns]
4   plan_type                            21 non-null     object
5   contract_type                        21 non-null     object
6   cancellation_date                     6 non-null      datetime64[ns]
7   cancellation_reason                   6 non-null      object
8   monthly_charges                       21 non-null     float64
9   cltv                                 21 non-null     int64
10  churn_score                           21 non-null     int64
dtypes: datetime64[ns](3), float64(1), int64(2), object(5)
memory usage: 1.9+ KB
```

```
In [29]: df_db_support.head()
```

Out[29]:

	customerid	complaint_date	escalations	csat_score	col_1	comment
0	0003-MKNFE	2024-08-28 00:00:00	N	60	None	service issue
1	0003-MKNFE	2024-08-28 00:00:00	Y	10	None	demaned refund
2	0013-EXCHZ	2024-01-20 00:00:00	Y	20	None	None
3	0013-MHZWF	2025-03-18 00:00:00	N	90	None	guidance to renew
4	0013-SMEOE	2024-11-01 00:00:00	N	30	None	None

```
In [30]: df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
#   Column            Non-Null Count  Dtype
---  -
0   customerid        9 non-null     object
1   complaint_date    9 non-null     object
2   escalations       9 non-null     object
3   csat_score        9 non-null     int64
4   col_1             0 non-null     object
5   comment           4 non-null     object
dtypes: int64(1), object(5)
memory usage: 564.0+ bytes
```

```
In [31]: df_db_support.drop(columns=['col_1', 'comment'], inplace=True)
```

```
In [32]: df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      9 non-null     object
1   complaint_date  9 non-null     object
2   escalations     9 non-null     object
3   csat_score      9 non-null     int64
dtypes: int64(1), object(3)
memory usage: 420.0+ bytes
```

```
In [33]: df_db_support['complaint_date']=df_db_support['complaint_date'].apply(pd.to_datetime)
```

```
In [34]: df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      9 non-null     object
1   complaint_date  9 non-null     datetime64[ns]
2   escalations     9 non-null     object
3   csat_score      9 non-null     int64
dtypes: datetime64[ns](1), int64(1), object(2)
memory usage: 420.0+ bytes
```

Feature Engineering And Data Analysis

```
In [35]: df_db_subscription.head(3)
```

Out[35]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	A
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	A
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	M

```
In [36]: df_db_subscription['Churn Flag']= np.where(df_db_subscription['cancellation_date'].
```

```
In [37]: df_db_subscription['Churn Flag']
```

```
Out[37]: 0      0
          1      1
          2      0
          3      0
          4      1
          5      0
          6      1
          7      0
          8      0
          9      0
         10      0
         11      1
         12      0
         13      1
         14      0
         15      0
         16      0
         17      0
         18      1
         19      0
         20      0
          Name: Churn Flag, dtype: int32
```

```
In [38]: df_db_subscription
```

Out[38]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	N
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	N
5	0013-MHZWF	2022-06-18	Paid	2025-06-18	Standard	
6	0013-SMEOE	2021-09-30	Refferal	2024-09-30	Basic	N
7	0014-BMAQU	2020-02-14	Organic	2025-02-14	Premium	
8	0015-UOCOJ	2023-07-22	Organic	2024-07-22	Standard	N
9	0016-QLJIS	2022-04-03	Organic	2025-04-03	Basic	
10	0017-DINOC	2021-12-01	Organic	2025-12-01	Premium	
11	0017-IUDMW	2023-03-17	Refferal	2024-03-17	Standard	N
12	0018-NYROU	2020-10-09	Organic	2025-10-09	Standard	
13	0019-EFAEP	2022-08-25	Refferal	2024-08-25	Basic	N
14	0019-GFNTW	2019-11-11	Paid	2025-11-11	Premium	
15	0020-INWCK	2023-05-06	Organic	2025-05-06	Standard	N
16	0020-JDNXP	2022-01-19	Organic	2024-01-19	Premium	
17	0021-IKXGC	2021-07-07	Paid	2025-07-07	Standard	
18	0022-TCJCI	2023-09-14	Refferal	2024-09-14	Basic	N
19	0023-HGHWL	2020-06-23	Organic	2025-06-23	Premium	

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
20	0023-UYUPN	2022-12-31	Paid	2025-12-31	Standard	N

```
In [68]: df=(df_db_subscription
            .merge(df_db_customer, on='customerid',how='left')
            .merge(df_db_support, on='customerid',how='left'))
```

```
In [69]: df
```

Out[69]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	N
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	N
5	0013-MHZWF	2022-06-18	Paid	2025-06-18	Standard	
6	0013-SMEOE	2021-09-30	Refferal	2024-09-30	Basic	N
7	0014-BMAQU	2020-02-14	Organic	2025-02-14	Premium	
8	0015-UOCOJ	2023-07-22	Organic	2024-07-22	Standard	N
9	0016-QLJIS	2022-04-03	Organic	2025-04-03	Basic	
10	0017-DINOC	2021-12-01	Organic	2025-12-01	Premium	
11	0017-IUDMW	2023-03-17	Refferal	2024-03-17	Standard	N
12	0018-NYROU	2020-10-09	Organic	2025-10-09	Standard	
13	0019-EFAEP	2022-08-25	Refferal	2024-08-25	Basic	N
14	0019-GFNTW	2019-11-11	Paid	2025-11-11	Premium	
15	0020-INWCK	2023-05-06	Organic	2025-05-06	Standard	N
16	0020-JDNXP	2022-01-19	Organic	2024-01-19	Premium	
17	0021-IKXGC	2021-07-07	Paid	2025-07-07	Standard	
18	0022-TCJCI	2023-09-14	Refferal	2024-09-14	Basic	N

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
19	0023-HGHWL	2020-06-23	Organic	2025-06-23	Premium	
20	0023-UYUPN	2022-12-31	Paid	2025-12-31	Standard	N

21 rows × 21 columns

In [70]: `df.shape`

Out[70]: (21, 21)

In [71]: `df_db_customer.shape`

Out[71]: (21, 6)

In [72]: `df_db_subscription.shape`

Out[72]: (21, 12)

In [73]: `df_db_support.shape`

Out[73]: (7, 5)

In [74]: `df_db_support`

Out[74]:

	customerid	complaint_date	escalations	csat_score	complaints_count
2	0013-EXCHZ	2024-01-20	Y	20	1
5	0017-IUDMW	2024-04-10	Y	25	1
1	0003-MKNFE	2024-08-28	Y	10	1
8	0022-TCJCI	2024-09-14	N	90	1
6	0019-EFAEP	2024-09-27	Y	30	1
4	0013-SMEOE	2024-11-01	N	30	1
3	0013-MHZWF	2025-03-18	N	90	1

In [79]: `df=(df_db_subscription
 .merge(df_db_customer, on='customerid',how='left')
 .merge(df_db_support, on='customerid',how='left'))`

In [75]: `df_db_support['complaints_count']=df_db_support.groupby('customerid')['customerid'].`

In [76]: `df_db_support`

Out[76]:

	customerid	complaint_date	escalations	csat_score	compaints_count
2	0013-EXCHZ	2024-01-20	Y	20	1
5	0017-IUDMW	2024-04-10	Y	25	1
1	0003-MKNFE	2024-08-28	Y	10	1
8	0022-TCJCI	2024-09-14	N	90	1
6	0019-EFAEP	2024-09-27	Y	30	1
4	0013-SMEOE	2024-11-01	N	30	1
3	0013-MHZWF	2025-03-18	N	90	1

In [77]: `df_db_support=df_db_support.sort_values('complaint_date').drop_duplicates('customer`

In [78]: `df_db_support`

Out[78]:

	customerid	complaint_date	escalations	csat_score	compaints_count
2	0013-EXCHZ	2024-01-20	Y	20	1
5	0017-IUDMW	2024-04-10	Y	25	1
1	0003-MKNFE	2024-08-28	Y	10	1
8	0022-TCJCI	2024-09-14	N	90	1
6	0019-EFAEP	2024-09-27	Y	30	1
4	0013-SMEOE	2024-11-01	N	30	1
3	0013-MHZWF	2025-03-18	N	90	1

In [50]: `df.to_csv('exported_churn_data.csv',index=False)`

In []:

Data Analysis

In [51]: `df.columns`

Out[51]: Index(['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'Churn Flag', 'Customer_Name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score'], dtype='object')

In [52]: `churn_rate = df['Churn Flag'].mean()*100
print('Churn Rate ',round(churn_rate,2),'%')`

Churn Rate 34.78 %

```
In [53]: retention_rate = 100-churn_rate
print('Retention Rate ',round(retention_rate,2), '%')
```

Retention Rate 65.22 %

```
In [54]: churn_by_plan = df.groupby('plan_type')['Churn Flag'].mean().mul(100).round(2).reset_index()
print(churn_by_plan)
```

	plan_type	churn_rate_pct
0	Basic	66.67
1	Premium	25.00
2	Standard	22.22

```
In [55]: # churn by rate + sum(revenue) & count(users)
# churn by subscription type
```

```
In [56]: churn_by_state = df.groupby('state')['Churn Flag'].mean().mul(100).round(2)
print(churn_by_state)
```

state	
Delhi	25.00
Karnataka	100.00
Kathmandu	0.00
Maharashtra	0.00
Meghalaya	66.67
Nagaland	0.00
Rajasthan	0.00
Telangana	66.67
Uttar Pradesh	0.00

Name: Churn Flag, dtype: float64

```
In [57]: # ARPU- Average Revenue per user
arpu = df['monthly_charges'].mean()
print('ARPU : ',round(arpu,2))
```

ARPU : 18.51

```
In [58]: # Average Customer Tenure
Today = pd.Timestamp.today()
df['tenure_days'] = np.where(
    df['cancellation_date'].notna(),
    (df['cancellation_date']-df['subscription_start_date']).dt.days,
    (Today-df['subscription_start_date']).dt.days
)
df.head()
```

Out[58]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	A
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	A
2	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	A
3	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	M
4	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	A

5 rows × 21 columns



In [59]: `df.columns`

Out[59]: Index(['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'Churn Flag', 'Customer_Name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'tenure_days'], dtype='object')

In [60]: `average_tenure = df['tenure_days'].mean()`
`print('Average Tenure(Days): ',round(average_tenure,0))`

Average Tenure(Days): 1499.0

In [130... `df.head()`

Out[130...

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	A
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	A
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	M
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	A
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	M

5 rows × 22 columns



In [131... `df.columns`

Out[131... Index(['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'Churn Flag', 'Customer_Name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'complaints_count', 'churn_risk'], dtype='object')

In [132... `# Revenue at risk - revenue lost from churn uers`
`revenue_at_risk = df.loc[df['Churn Flag']==1,'monthly_charges'].sum()`
`print('Revenue at Risk (K) = ', revenue_at_risk)`

Revenue at Risk (K) = 73.94

In [133... `escalations_rate = (df['escalations']=='Y').mean()*100`
`print('Escalations Rate: ',round(escalations_rate,2),'%')`

Escalations Rate: 19.05 %

In [134... `# Average Complaint Per User`
`avg_complaints = df['compaints_count'].sum()/df['customerid'].nunique()`
`print('Avg Complaint Per User =',round(avg_complaints,2))`

Avg Complaint Per User = 0.33

In [135... `df['escalations'] = np.where(df['escalations']=='Y',1,0)`

In [136... `corr_df = df[['escalations','Churn Flag']].dropna()`
`correlation = corr_df['escalations'].corr(df['Churn Flag'])`
`print('Correlation between Esclation vs Churn Flag =', round(correlation,2))`

Correlation between Esclation vs Churn Flag = 0.77

In [137... `condtions = [`
`(df['churn_score']<50),`

```
(df['churn_score']>=50)&(df['churn_score']<70),  
(df['churn_score']>=70)  
]  
choices = ['Low','Med','High']  
df['churn_risk']=np.select(conditions,choices,default='unknown')
```

```
In [138... df[['churn_risk','churn_score']].head()
```

```
Out[138...  


|   | churn_risk | churn_score |
|---|------------|-------------|
| 0 | Low        | 12          |
| 1 | High       | 91          |
| 2 | Low        | 34          |
| 3 | Low        | 8           |
| 4 | High       | 88          |


```

```
In [ ]:
```

```
In [139... df_visual = df.copy()
```

```
In [140... df_visual
```

Out[140...

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	N
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	N
5	0013-MHZWF	2022-06-18	Paid	2025-06-18	Standard	
6	0013-SMEOE	2021-09-30	Refferal	2024-09-30	Basic	N
7	0014-BMAQU	2020-02-14	Organic	2025-02-14	Premium	
8	0015-UOCOJ	2023-07-22	Organic	2024-07-22	Standard	N
9	0016-QLJIS	2022-04-03	Organic	2025-04-03	Basic	
10	0017-DINOC	2021-12-01	Organic	2025-12-01	Premium	
11	0017-IUDMW	2023-03-17	Refferal	2024-03-17	Standard	N
12	0018-NYROU	2020-10-09	Organic	2025-10-09	Standard	
13	0019-EFAEP	2022-08-25	Refferal	2024-08-25	Basic	N
14	0019-GFNTW	2019-11-11	Paid	2025-11-11	Premium	
15	0020-INWCK	2023-05-06	Organic	2025-05-06	Standard	N
16	0020-JDNXP	2022-01-19	Organic	2024-01-19	Premium	
17	0021-IKXGC	2021-07-07	Paid	2025-07-07	Standard	
18	0022-TCJCI	2023-09-14	Refferal	2024-09-14	Basic	N

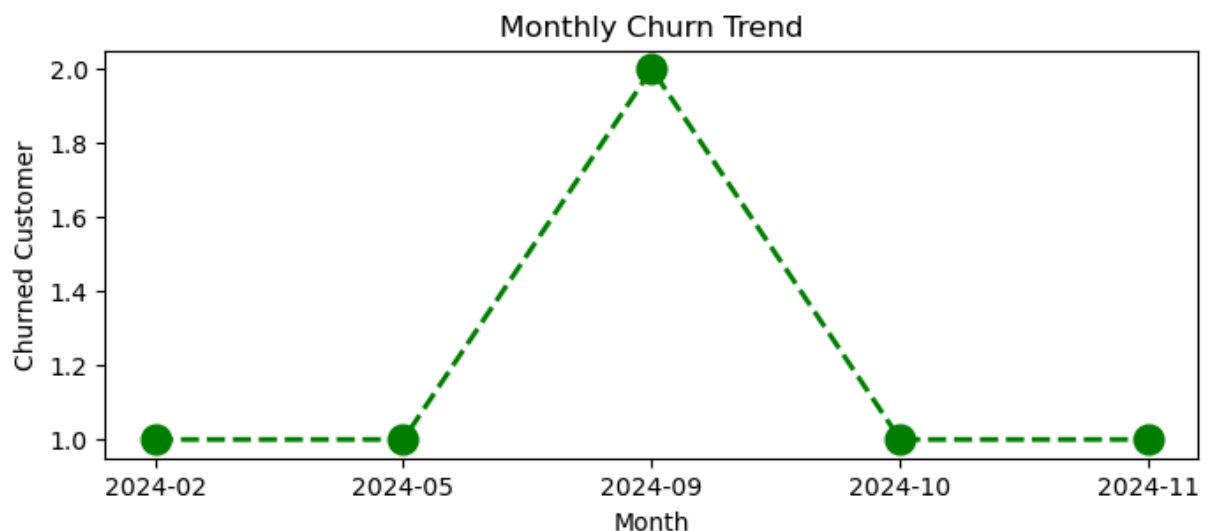
	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract_type
19	0023-HGHWL	2020-06-23	Organic	2025-06-23	Premium	
20	0023-UYUPN	2022-12-31	Paid	2025-12-31	Standard	M

21 rows × 22 columns

In [141...] `df_visual.columns`

Out[141...] `Index(['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score', 'Churn Flag', 'Customer_Name', 'country', 'state', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'complaints_count', 'churn_risk'], dtype='object')`

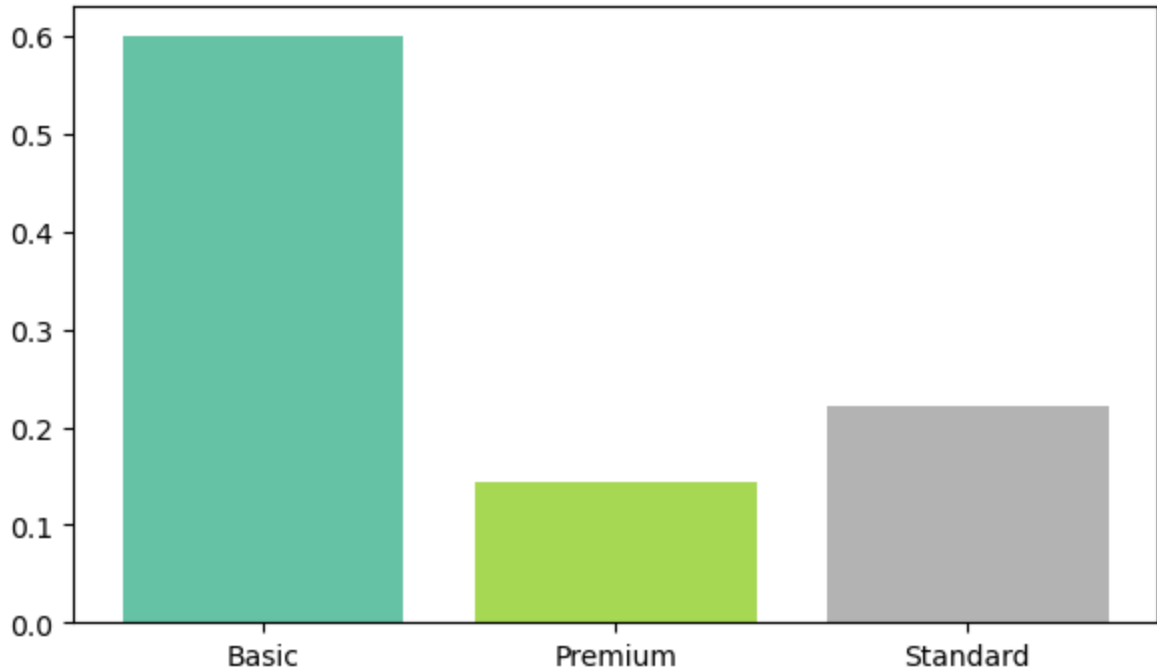
In [142...] `df_visual['cancellation_month']=df_visual['cancellation_date'].dt.to_period('M')`
`churn_trend = df_visual[df_visual['Churn Flag']==1].groupby('cancellation_month').size()`
`plt.figure(figsize=(8,3))`
`plt.plot(churn_trend.index.astype(str), churn_trend.values,color='green', marker='o')`
`plt.title('Monthly Churn Trend')`
`plt.xlabel('Month')`
`plt.ylabel('Churned Customer')`
`plt.show()`



In [143...] `churn_plan = df_visual.groupby('plan_type')['Churn Flag'].mean()`
`# colors = ['yellow', 'red', 'blue']`
`colors = plt.cm.Set2(np.linspace(0,1,len(churn_plan)))`

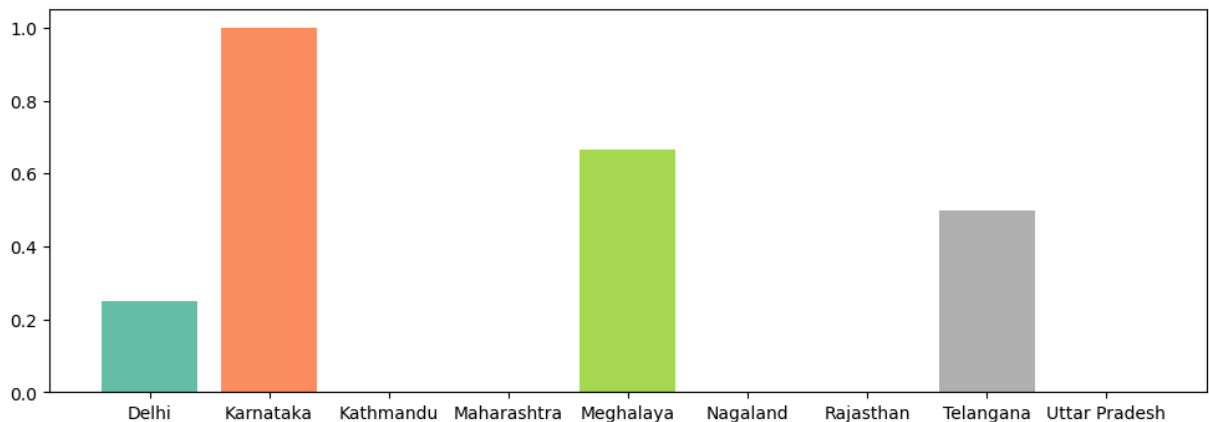
`plt.figure(figsize=(7,4))`

```
plt.bar(churn_plan.index,churn_plan.values, color = colors)
plt.show()
```



```
In [144...] churn_state = df_visual.groupby('state')['Churn Flag'].mean()
# colors = ['yellow', 'red', 'blue']
colors = plt.cm.Set2(np.linspace(0,1,len(churn_state)))

plt.figure(figsize=(12,4))
plt.bar(churn_state.index,churn_state.values, color = colors)
plt.show()
```



```
In [145...] import warnings
warnings.filterwarnings('ignore')
```

```
In [146...] # incorrect method of encoding - as numbers are not assigned based on priority
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'Churn Flag',
categorical_cols = ['plan_type', 'contract_type', 'churn_risk']
for col in categorical_cols:
    df_encoded[col]=df_encoded[col].astype('category').cat.codes
```

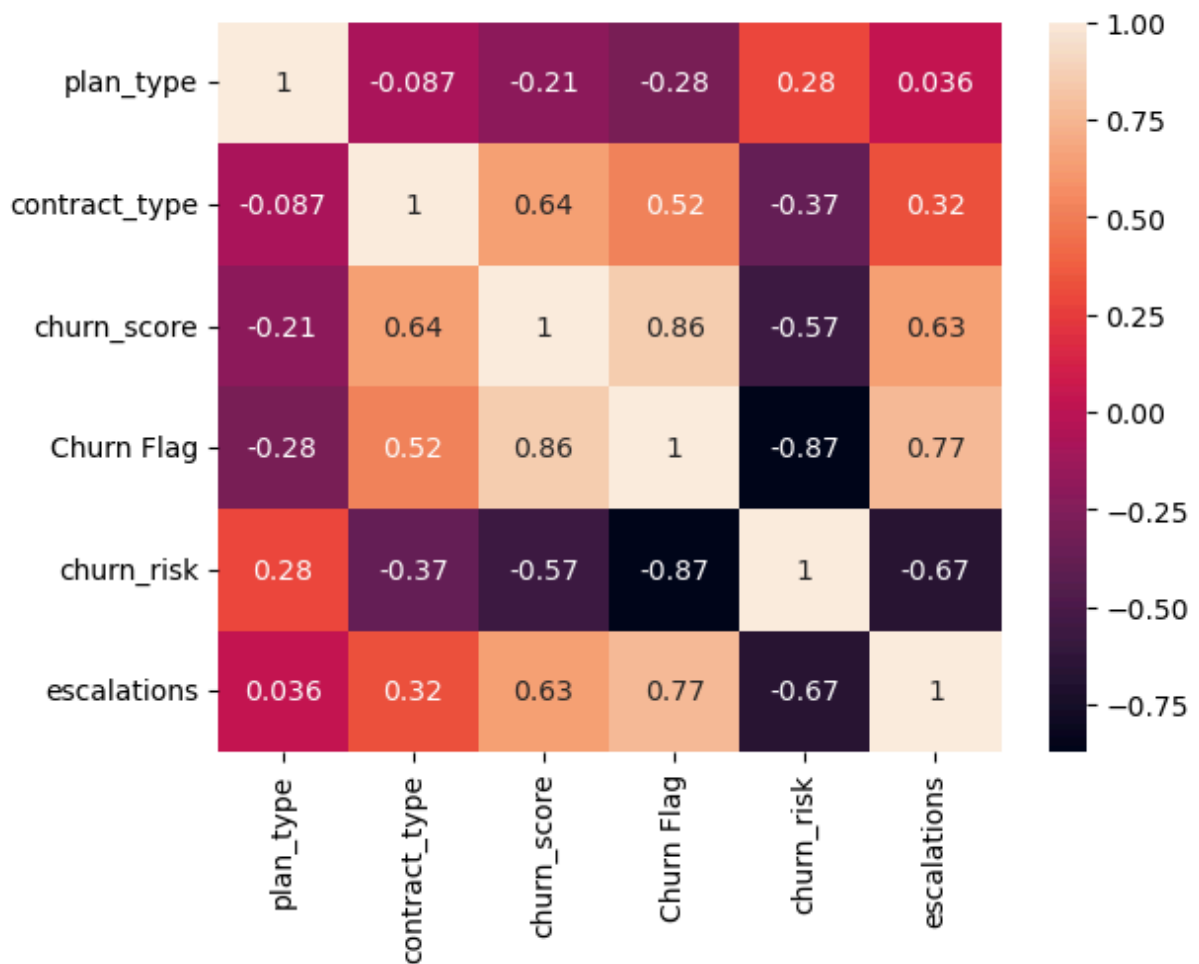
```
In [147... df_encoded.head()
```

```
Out[147...
```

	plan_type	contract_type	churn_score	Churn Flag	churn_risk	escalations
0	2	0	12	0	1	0
1	1	0	91	1	0	1
2	0	1	34	0	1	0
3	1	0	8	0	1	0
4	2	1	88	1	0	1

```
In [148... sns.heatmap(df_encoded.corr(),annot=True)
```

```
Out[148... <Axes: >
```



```
In [156... df_visual['plan_type'].unique()
```

```
Out[156... array(['Standard', 'Premium', 'Basic'], dtype=object)
```

```
In [157... # Correct method of encoding - based on priority
df_encoded = df_visual[['plan_type','contract_type','churn_score','Churn Flag','churn_risk','escalations']]
order_mapping = {'plan_type':['Basic','Standard','Premium'],
```

```

        'contract_type':['Monthly','Annual'],
        'churn_risk':['Low','Med','High']
    }
    for col, order in order_mapping.items():
        df_encoded[col]=pd.Categorical(df_encoded[col].astype('category'),categories=or

```

In [160...] `df_visual[['plan_type','contract_type','churn_score','Churn Flag','churn_risk','esc`

Out[160...]

	plan_type	contract_type	churn_score	Churn Flag	churn_risk	escalations
0	Standard	Annual	12	0	Low	0
1	Premium	Annual	91	1	High	1
2	Basic	Monthly	34	0	Low	0
3	Premium	Annual	8	0	Low	0
4	Standard	Monthly	88	1	High	1

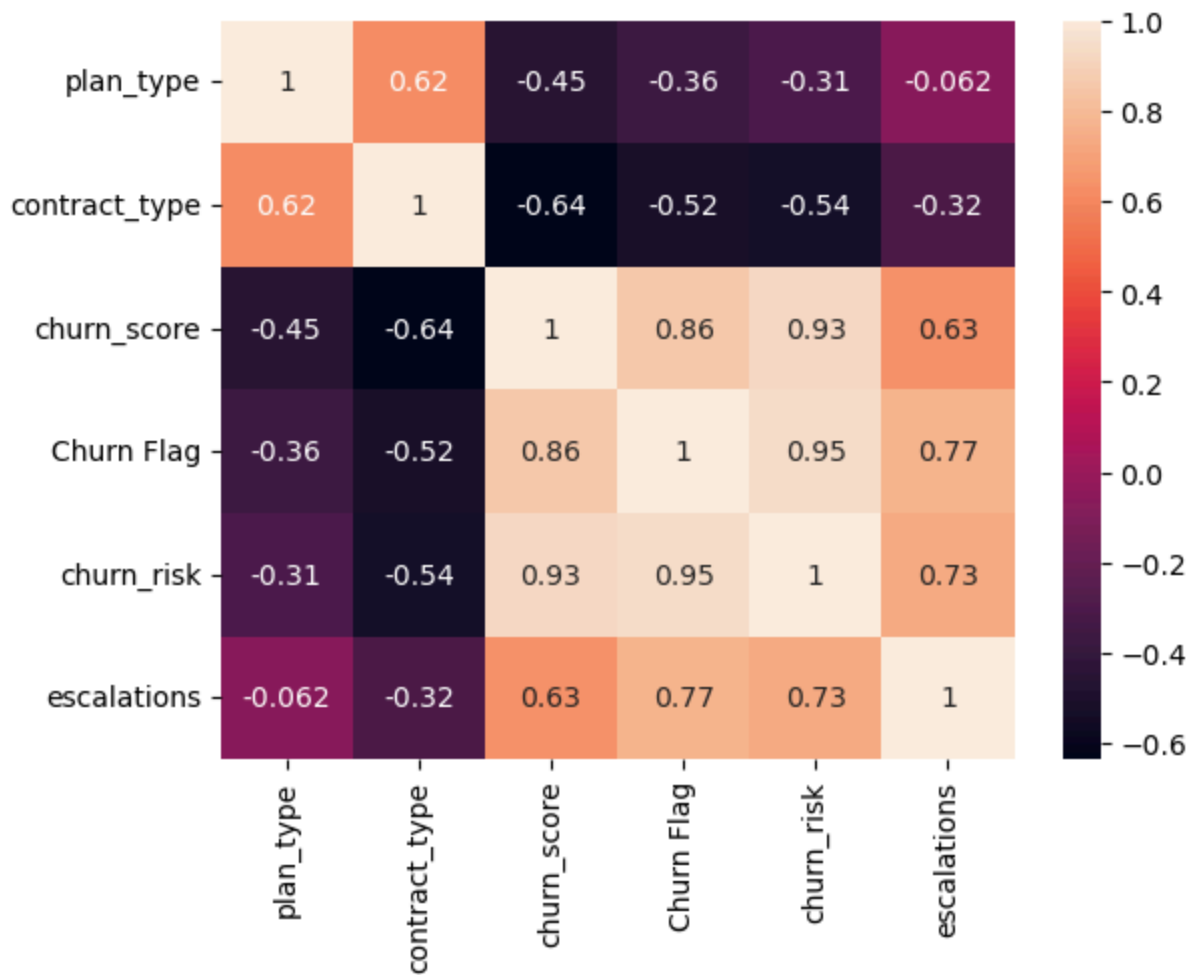
In [158...] `df_encoded.head()`

Out[158...]

	plan_type	contract_type	churn_score	Churn Flag	churn_risk	escalations
0	1	1	12	0	0	0
1	2	1	91	1	2	1
2	0	0	34	0	0	0
3	2	1	8	0	0	0
4	1	0	88	1	2	1

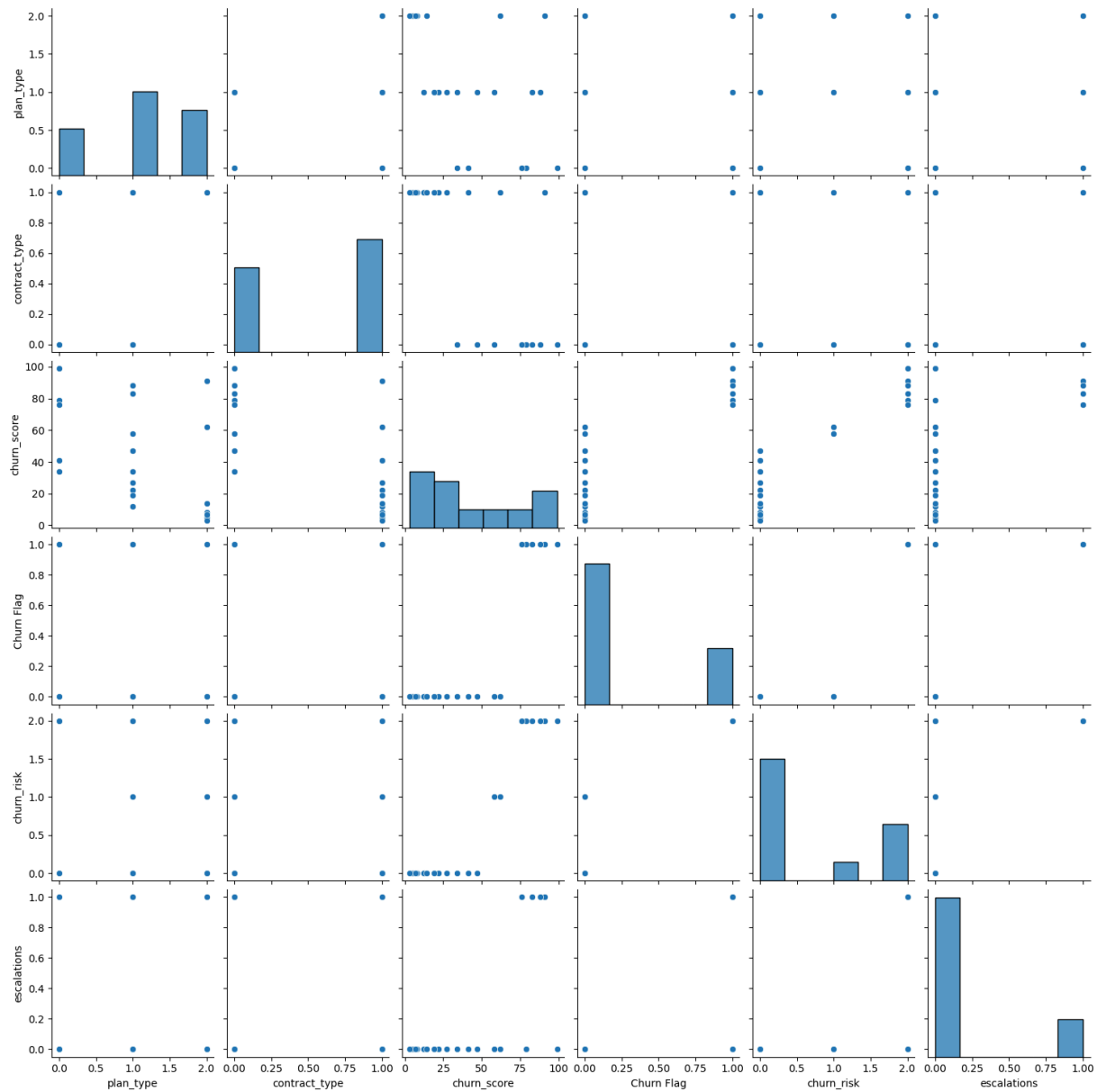
In [161...] `sns.heatmap(df_encoded.corr(),annot=True)`

Out[161...] `<Axes: >`



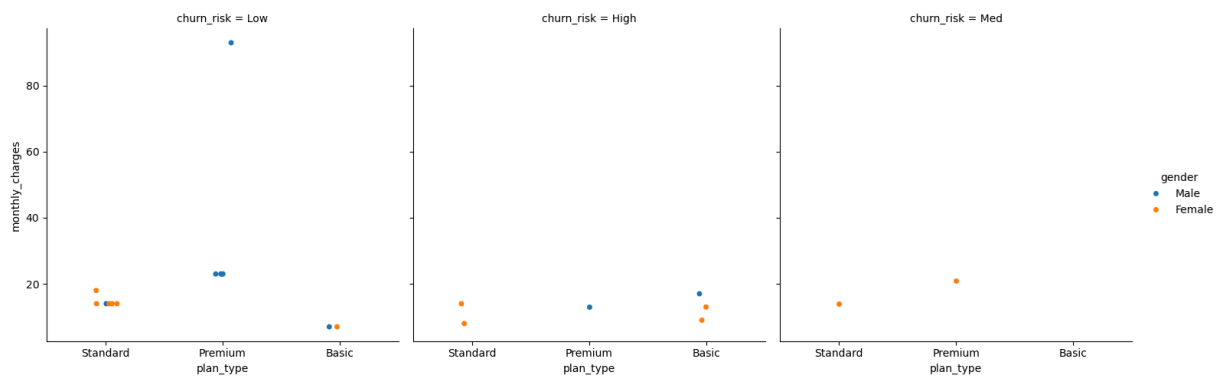
```
In [166... # PairPlot - relationship in a dataset
sns.pairplot(df_encoded)
```

```
Out[166... <seaborn.axisgrid.PairGrid at 0x16313e834a0>
```



```
In [165... # catplt/facegrid plot - multi dim comparison
sns.catplot(data = df_visual,
            x='plan_type',
            y='monthly_charges',
            hue='gender',
            col='churn_risk'
            )
```

Out[165... <seaborn.axisgrid.FacetGrid at 0x16311d2baa0>



In [170...

```
# Pivot Table
pd.pivot_table(
    df_visual,
    values='Churn Flag',
    index='plan_type',
    aggfunc='mean'
)
```

Out[170...

Churn Flag	
plan_type	
Basic	0.600000
Premium	0.142857
Standard	0.222222

In []: